

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2023 Course: B. Tech. Branch: Mechanical Engineering / Mechanical Engineering (Sandwich) / Production Engineering Semester: VIII Subject Code & Name: BTMEC802F & Non-Conventional Energy Resources Max Marks: 60 Date:17-01-24 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	List the advantages and disadvantages of non-conventional energy sources		6
B)	What are energy resources available in India? Explain		6
C)	What is fossil fuel? What are different alternatives for fossil fuels?		6
Q.2	Solve Any Two of the following.		12
A)	What is a need of alternative source? Explain by considering solar energy		6
B)	With neat sketch explain the working of Solar pond.		6
C)	Enumerate the different types of concentrating type collectors		6
Q. 3	Solve Any Two of the following.		12
A)	What is the basic principle of ocean thermal energy conversion? What are the main types of OTEC power plants?		6
B)	What factors affect the performance of solar flat plate collector?		6
C)	How can communities promote the adoption of solar energy and renewable practices?		6
Q.4	Solve Any Two of the following.		12
A)	Describe the functions of components of wind energy systems		6
B)	What are the merits and demerits of geothermal energy?		6

C)	What is tide? Explain the basic components of a tidal power plant and state their merits and demerits.		6
Q. 5	Solve Any Two of the following.		12
A)	What is the geothermal energy? Explain its extraction process.		6
B)	List and explain the challenges associated with Geothermal energy		6
C)	Explain the working principle of MHD power generation		6
	*** End ***		

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